

# RoboBlast Grenade Verifier

Status: PASS  
Godot: 4.6-stable (official)  
Generated: 2026-07-03 22:31  
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Source: ...t-runs-20260703-151656\r  
un-03-codex\evidence\score.json

Executive score report for deterministic grenade weapon benchmark output.

100/100

PASS

100 / 100

Total benchmark score

Passing threshold met.

The PDF mirrors the verifier JSON result. Category bars below show which behaviors were observed at runtime and which remain incomplete.

## Category Scores

| Category                | Score | Observed    |
|-------------------------|-------|-------------|
| Weapon Controls         | 15/15 | <div></div> |
| Hud Feedback            | 10/10 | <div></div> |
| Trajectory Preview      | 30/30 | <div></div> |
| Projectile Physics      | 15/15 | <div></div> |
| Explosion Gameplay      | 20/20 | <div></div> |
| Visual Audio Polish     | 5/5   | <div></div> |
| Stability Repeatability | 5/5   | <div></div> |

## Key Findings

|                               |                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Weapon Controls (15/15)    | a weapon-switch input route also carries a controller binding; weapon mode observably switches through weapon_switch action; attack after weapon switch created runtime nodes; rapid grenade attacks appear rate-limited; default aimed attack still creates runtime nodes                                          |
| 2. Hud Feedback (10/10)       | player has visible UI controls; UI control state changed after weapon switch; aiming changes UI feedback                                                                                                                                                                                                            |
| 3. Trajectory Preview (30/30) | visible grenade aiming aid appears in grenade mode; aiming aid communicates an arcing throw or landing area; aiming aid updates after aim or camera direction changes; aiming aid direction matches the thrown grenade direction; aiming aid remains available in grenade mode and hides after leaving grenade mode |
| 4. Projectile Physics (15/15) | grenade attack spawned a nearby 3D node; spawned node follows arcing motion; projectile path stayed clear of player body                                                                                                                                                                                            |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5. Explosion Gameplay (20/20)</b> | default throw calibration full at 10.34 units: default throw calibration measured 10.34 units from GrenadeProjectile after 8.69 units of horizontal travel; nearby target damage averaged across explosion trials: Seed 17031 throw raw nearby hit score 10/10; Seed 27059 throw raw nearby hit score 10/10; Seed 37087 throw raw nearby hit score 10/10; nearby damageable-only destructible damage averaged across explosion trials: Seed 17031 throw nearby damageable-only destructible was damaged; Seed 27059 throw nearby damageable-only destructible was damaged; Seed 37087 throw nearby damageable-only destructible was damaged; blast locality averaged across explosion trials: Seed 17031 throw explosion damage stayed localized; Seed 27059 throw explosion damage stayed localized; Seed 37087 throw explosion damage stayed localized; all explosion safety trials protected out-of-range targets; player safety averaged across explosion trials: Seed 17031 throw player safe; Seed 27059 throw player safe; Seed 37087 throw player safe; detonation effects averaged across explosion trials: Seed 17031 throw detonation produced runtime nodes or effects; Seed 27059 throw detonation produced runtime nodes or effects; Seed 37087 throw detonation produced runtime nodes or effects; default throw calibration is full quality inside the 6-12 unit envelope |
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## Detailed Item Analysis

### Weapon Controls Detail (15/15)

| Result            | Check                            | Evidence                                                      |
|-------------------|----------------------------------|---------------------------------------------------------------|
| <b>Earned 2/2</b> | Controller weapon-switch binding | a weapon-switch input route also carries a controller binding |
| <b>Earned 2/2</b> | Weapon switch input responds     | weapon mode observably switches through weapon_switch action  |
| <b>Earned 5/5</b> | Grenade attack after switching   | attack after weapon switch created runtime nodes              |
| <b>Earned 3/3</b> | Rapid attack rate limit          | rapid grenade attacks appear rate-limited                     |
| <b>Earned 3/3</b> | Default weapon still works       | default aimed attack still creates runtime nodes              |

### Hud Feedback Detail (10/10)

| Result            | Check                  | Evidence                                     |
|-------------------|------------------------|----------------------------------------------|
| <b>Earned 3/3</b> | Visible UI controls    | player has visible UI controls               |
| <b>Earned 4/4</b> | Weapon-switch UI state | UI control state changed after weapon switch |
| <b>Earned 3/3</b> | Aiming UI feedback     | aiming changes UI feedback                   |

### Trajectory Preview Detail (30/30)

| Result            | Check                                  | Evidence                                                                          |
|-------------------|----------------------------------------|-----------------------------------------------------------------------------------|
| <b>Earned 5/5</b> | Visible grenade aiming aid             | visible grenade aiming aid appears in grenade mode                                |
| <b>Earned 6/6</b> | Communicates arcing throw              | aiming aid communicates an arcing throw or landing area                           |
| <b>Earned 8/8</b> | Updates with aim/camera direction      | aiming aid updates after aim or camera direction changes                          |
| <b>Earned 7/7</b> | Preview matches projectile direction   | aiming aid direction matches the thrown grenade direction                         |
| <b>Earned 4/4</b> | Visibility lifecycle/cooldown behavior | aiming aid remains available in grenade mode and hides after leaving grenade mode |

### Projectile Physics Detail (15/15)

| Result            | Check              | Evidence                                    |
|-------------------|--------------------|---------------------------------------------|
| <b>Earned 4/4</b> | Projectile spawned | grenade attack spawned a nearby 3D node     |
| <b>Earned 8/8</b> | Arcing motion      | spawned node follows arcing motion          |
| <b>Earned 3/3</b> | Player-safe path   | projectile path stayed clear of player body |

**Explosion Gameplay Detail (20/20)**

| Result            | Check                                    | Evidence                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Earned 5/5</b> | Nearby target damage across angles       | default throw calibration full at 10.34 units: default throw calibration measured 10.34 units from GrenadeProjectile after 8.69 units of horizontal travel; nearby target damage averaged across explosion trials: Seed 17031 throw raw nearby hit score 10/10; Seed 27059 throw raw nearby hit score 10/10; Seed 37087 throw raw nearby hit score 10/10 |
| <b>Earned 3/3</b> | Nearby destructible damage across angles | nearby damageable-only destructible damage averaged across explosion trials: Seed 17031 throw nearby damageable-only destructible was damaged; Seed 27059 throw nearby damageable-only destructible was damaged; Seed 37087 throw nearby damageable-only destructible was damaged                                                                        |
| <b>Earned 5/5</b> | Blast locality across angles             | blast locality averaged across explosion trials: Seed 17031 throw explosion damage stayed localized; Seed 27059 throw explosion damage stayed localized; Seed 37087 throw explosion damage stayed localized; all explosion safety trials protected out-of-range targets                                                                                  |
| <b>Earned 2/2</b> | Player safety across angles              | player safety averaged across explosion trials: Seed 17031 throw player safe; Seed 27059 throw player safe; Seed 37087 throw player safe                                                                                                                                                                                                                 |
| <b>Earned 3/3</b> | Detonation effects across angles         | detonation effects averaged across explosion trials: Seed 17031 throw detonation produced runtime nodes or effects; Seed 27059 throw detonation produced runtime nodes or effects; Seed 37087 throw detonation produced runtime nodes or effects                                                                                                         |
| <b>Earned 2/2</b> | Throw distance calibration quality       | default throw calibration is full quality inside the 6-12 unit envelope                                                                                                                                                                                                                                                                                  |

**Visual Audio Polish Detail (5/5)**

| Result            | Check                  | Evidence                                         |
|-------------------|------------------------|--------------------------------------------------|
| <b>Earned 2/2</b> | Visible effect nodes   | visible grenade or explosion nodes appeared      |
| <b>Earned 2/2</b> | Detonation audio       | audio player was active during detonation window |
| <b>Earned 1/1</b> | Temporary node cleanup | some temporary nodes cleaned up                  |

**Stability Repeatability Detail (5/5)**

| Result            | Check                            | Evidence                                                                                    |
|-------------------|----------------------------------|---------------------------------------------------------------------------------------------|
| <b>Earned 1/1</b> | Repeated grenade attacks         | two separated grenade attacks produced runtime behavior                                     |
| <b>Earned 1/1</b> | Default attack after grenade use | default aimed attack still works after grenade use                                          |
| <b>Earned 1/1</b> | Main scene loads                 | main scene loaded with a playable player                                                    |
| <b>Earned 1/1</b> | Main scene default attacks       | main scene default shooting and melee both produced runtime behavior                        |
| <b>Earned 1/1</b> | Main scene actors and pickups    | main scene kept targeteables, damageables, collect_coin, and damage-spawned pickup behavior |